

Project Proposal for Data Visualization

Investigating the Shift in European Car Registrations

Gianluca Rigatti - ID 265303

Giuseppe Screnci - ID 263989

1 Project Idea Description

1.1 Topic

The project focuses on the evolution of vehicle registrations by fuel type, with particular attention to the rise of electrified vehicles (battery electric, plug-in hybrid, and hybrid) compared to traditional internal combustion engine (ICE) vehicles (petrol and diesel) in the European car market.

We intend to address widespread claims that the general market shift to electrified power trains is primarily driven by national incentives and taxation rules [2]. We believe that a large portion of existing data visualizations and statistical analyses [3] ignore the fact that market offerings themselves are changing, not necessarily reflecting consumer demand.

Therefore, the central question guiding the analysis is:

"Is the observed increase in electrified vehicles driven by genuine growth in demand, national incentives and taxation rules, or is it largely a consequence of the decline in petrol and diesel vehicle offerings?"

1.2 Circumstances

1.2.1 Target Audience

This project aims to challenge common misconceptions found in existing data visualizations that focus on static market share breakdowns [1]. We feel such visualizations present a simplistic evolution of vehicle registrations by fuel types, failing to inform on the factors that are truly driving such shifts. As a result, a reader not well versed in the intricacies of the automotive market may conclude that electrified vehicles are genuinely gaining popularity and are preferred over ICE alternatives, potentially misinterpreting shifts in market composition as shifts in consumer preference.

Since these types of visualizations are primarily aimed at informing the general audience, this project adopts the same target, while seeking to provide a more informative and nuanced perspective.

1.2.2 Format and Medium

The project will be implemented as an interactive dashboard, developed using the *Dash* or *Streamlit* frameworks in Python. We believe an interactive format is particularly suited to the task given the temporal nature of the data and the presence of multiple categories to be analysed.

Interactivity will allow users to explore different countries (for example, comparing Italy with the European Union aggregate), examine variations across fuel categories, and switch between absolute and normalized values using various techniques, where normalization will attempt to account for various factors across car models.

1.3 Purpose

The visualization aims to demonstrate that an increasing share of electrified vehicles does not necessarily correlate to a strong absolute growth in demand. Instead, it seeks to highlight how the decline of petrol and diesel vehicles is significantly contributing to the market shift that is taking place.

After interacting with the visualization, the audience should understand that the transition towards electrified vehicles is driven by a combination of actual growth in electrified vehicle demand and a structural decline in traditional fuel vehicles.

2 Project Data

The project will rely on the following public datasets:

- [Eurostat’s New passenger cars by type of motor energy and engine size](#) dataset, which provides information on Petrol and Diesel vehicle registrations across European countries, broken down by engine size and time period.
- [Eurostat’s New passenger cars by type of motor energy](#) dataset, which provides information on vehicle registrations across European countries, broken down by fuel type and time period.
- [UNECE’s New passenger car registrations by fuel type \(detailed\)](#) dataset, which also provides information on vehicle registrations across European countries broken down by engine size and time period; this dataset aggregates the first two datasets covering a longer historical time span.
- [European Environment Agency \(EEA\)’s Monitoring of CO2 emissions from passenger cars Regulation \(EU\) 2019/631](#) dataset, which provides information on individual vehicle models registered across European countries, detailing their specific fuel types and physical specifications (such as emissions, mass, wheelbase, and engine power) over time.

These data sources, together, should enable the comparison between demand-side dynamics (registrations) and supply-side evolution (availability of different fuel types).

Given the datasets come from official European institutions and other international institutions we believe they provide a trustworthy foundation for us to build our analysis on.

To obtain the data in the format we need, significant preprocessing is required. This includes grouping fuel type classifications and processing the EEA dataset in order to identify unique vehicle models and their respective fuel type variants. We will also have to carefully consider which variables will be best suited to be used as input features for our deep learning model responsible for learning a meaningful lower dimensional representation for the chosen features, which will allow us to analyse just how broad the market offerings truly are for every power train type.

Some limitations must be acknowledged:

- The lack of a general year-by-year vehicle availability dataset makes the analysis difficult.
- Model availability is not directly observable. For instance, if a consumer prefers an ICE vehicle but faces limited availability, he may instead opt for an electrified alternative, making it difficult to interpret observed consumer choices and attribute them to preference.
- As highlighted in the cited articles, national incentives and taxation rules promoting electrification can significantly influence purchasing decisions, shifting the question from a matter of preference to a matter of convenience.

Despite these limitations, we believe the analysis will provide meaningful insights into the underlying market dynamics.

3 Project Definition

3.1 Data Requirements

Given the previously mentioned lack of a general dataset regarding model availability, and the large size of the datasets involved in our analyses (~ 40 GB), preprocessing and dimensionality reduction will be necessary to achieve our goals.

First off, we have to quantify model availability by power train type to be able to understand how broad the choice available to the consumer is when purchasing a new vehicle. This helps us determine whether purchasing decisions reflect real preferences or are influenced by limited options.

We intend to apply two different normalizations to the registration numbers to better understand the market shift to electrified power trains:

- **Normalization by Attributes (Baseline):** As a baseline metric, we will group the EEA datasets by their attributes:
 - We can group by *vehicle names*: This approach is a baseline and it must be noted that it risks artificially inflating the number of available vehicles for vehicles that have a lot of different trims.

- We can group by *vehicle names* and *unique power train options*: This approach would solve the artificial inflation of vehicle numbers problem, however it is susceptible to the dataset quality and the cleaning process.

The aforementioned grouping approaches are necessary if we wish to be able to get a discrete metric of the models available to the consumer.

- **Autoencoder-Based Normalization:** To establish a metric of true market diversity, we will use an Autoencoder for non-linear dimensionality reduction. We will compress the physical specifications of the vehicles (e.g., CO2 emissions, wheelbase, engine power) into a low-dimensional (2D or 3D) continuous latent space. This compression allows us to calculate the geometric volume that each fuel type occupies in the latent space, which represents the true choice offered to consumers and filters out homologation redundancies.

It is worth noting that this approach does not necessarily reflect how the consumer subjectively views the market, as consumers also consider brand and design of a vehicle.

Moreover, the way we decide to compute the volume (or the type of shape we fit to the points in the latent space) will influence the results. For instance, if we have a sparse set of points, fitting a sphere to points that are distant across all dimensions would result in a large estimated volume, even though the underlying set of consumer choices remains limited.

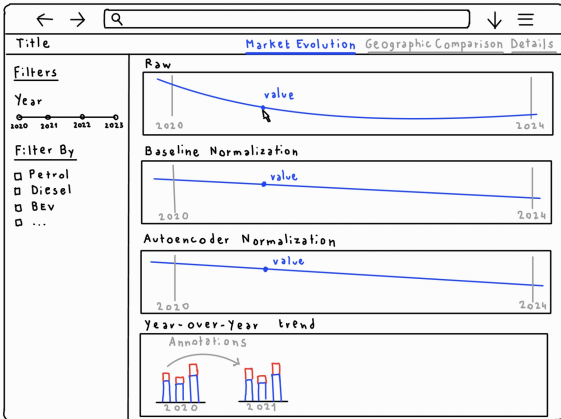
Training process and architecture: we will be using a small set of features as input to the network; we will gradually decrease the dimensionality through the network during the architectural design process to keep training stable until we reach our desired 2 or 3 dimensional representation. The training hyperparameters will be defined through an iterative process based on the various trained models' performance. As we only wish to reduce dimensionality with this method, the trained decoder will not be used in our solution.

3.2 Potential Views and Visual Encodings

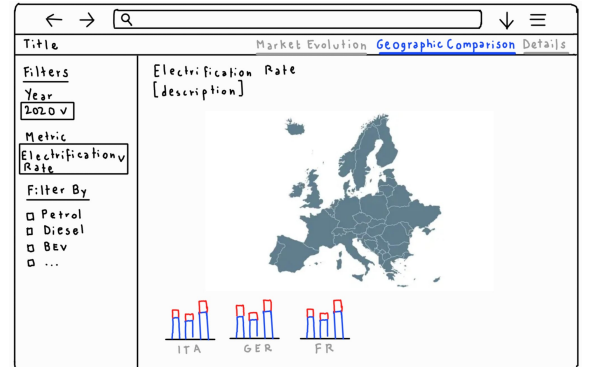
As shown in Figure 1a The core of the data visualization will consist of *time-series representations* to track the evolution of vehicle registrations per fuel type. In particular, three distinct trend lines will be presented for comparison: raw vehicle registration counts by fuel type, registrations normalized using a baseline measure of available models, and registrations normalized with respect to the estimated geometric volume of the Autoencoder's latent space.

To complement these data visualizations, *stacked bar charts* will illustrate how the composition of the market changes over time. These charts will emphasize the shifting balance between ICE and electrified vehicles when applying our normalization metrics.

Furthermore, as illustrated in Figure 1b a *comparative geographic view* across different European countries will highlight regional differences in adoption patterns, and thanks to the stacked bar charts the user will be able to see how the various normalisations affect different countries' numbers.



(a) A low fidelity wireframe for the time series part of our solution, meant to show the user the overall results of our analysis on the whole dataset(s), with year-over-year comparisons.



(b) A low fidelity wireframe of the geographic view planned for our solution, note the stacked bar charts representing raw and normalised data points for vehicle registration for the different fuel types.

Figure 1: Low fidelity wireframes for planned views in our solution.

Finally, the features used for training the Autoencoder will be reported in our solution, alongside a SHAP plot showing how heavily the Autoencoder relies on them to compute the value of specific latent dimensions. Color coding will be used to show whether high or low values for those features will push the latent dimension up or down. The geometric volume values for the market will be visualised in this same view. As per Subsection 1.2.1, these views will reside in a dedicated *details view* meant for a more technical audience.

3.3 Highlighting Key Features

Since the target audience is the general public, the visualization must be engaging and easy to understand. For this reason the design will adopt bright colors, thick lines and include clear explanations as to how results were computed. Such elements will help guide the user’s attention and reduce the risk of misinterpretation.

- Color will distinguish between ICE and electrified vehicles using consistent mappings across all views.
- Annotations will highlight significant declines or rapid increases in adoption or differences between the raw and normalized results.

The dashboard structure will follow a *narrative progression*. It will start with standard observed indicators, such as raw market share, and introduce the derived views. This progression guides the user through the methodology to support data-driven conclusions.

4 Expected Challenges

Extracting vehicle models and fuel types from the EEA dataset presents a technical challenge since identifying unique models requires addressing homologation redundancies, minor trim variations, and platform-sharing:

- The normalization based on model attributes relies heavily on the quality of the data cleaning phase. Inconsistencies in the raw data must be resolved before applying text-based grouping techniques to construct a reliable dataset. Any errors at this stage may directly affect the accuracy of the results.
- As we already mentioned in the Autoencoder-based normalization, we have to determine in which way we will compute the geometric volume, possible sparsity in the latent space will have to be taken into consideration; grid binning can be used to handle sparsity.
- The SHAP plot we will use to interpret how the Autoencoder weighs different features when computing latent space dimensions’ values may be visually overwhelming.

Interpreting the results requires caution. The Autoencoder approach quantifies physical utility but does not account for subjective consumer preferences regarding brand and aesthetics. There is a risk of overstating the causal relationship between the physical diversity of supply and vehicle registration, we can address this by framing our analyses as correlational and not strictly causal.

References

- [1] ACEA. New eu car sales by power source, 2024. URL: <https://www.acea.auto/figure/fuel-types-of-new-passenger-cars-in-eu/>.
- [2] Euronews. Petrol or electric: Evs are becoming more popular than fuel-powered cars in europe, 2026. URL: <https://www.euronews.com/my-europe/2026/02/02/petrol-or-electric-evs-are-becoming-more-popular-than-fuel-powered-cars-in-europe>.
- [3] Eurostat. Passenger cars in the eu, 2024. URL: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Passenger_cars_in_the_EU.